OPDI Papers

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Methodology studies behind the Open Performance Data Initiative.

Each paper benchmarks an OPDI algorithm against EUROCONTROL reference data and reports what the comparison actually showed, including where a proposed method failed. Every figure renders from a committed CSV cache — no credentials, no database, no cluster — so the results on any page can be reproduced from the repository alone.

1 Papers

Detecting departure and destination aerodromes from ADS-B — version 5 · PDF

Takes the arrival deficit apart. Finds it is silence rather than error, that most of that silence is the abstention refusing candidates it is right to refuse, and that a bearing test — does the trajectory point at the aerodrome? — is the first signal to beat it. Establishes why: a position difference survives the archive's carried-forward staleness, and a rate does not, which is why vertical rate, closing rate and the approach cone all failed.

Thinning state vectors, and what the archive actually contains · PDF

Tests whether OPDI's 5 s decimation rule — a fixed-phase `event_time % 5` filter — loses data where reception is intermittent. It does not: the OpenSky archive is a complete 1 Hz grid, and a correct bin-based sampler recovers 0.22% more rows. The experiment instead finds that the grid is padded with carried-forward positions, and that the staleness is wildly asymmetric — a departure's first fix is fresh, while an arrival's last fix carries a position measured a median of 190 seconds earlier.

Detecting departure and destination aerodromes from ADS-B — version 4.5 · PDF

Moves detection into the OPDI pipeline itself: H3 cells as the index, exact great-circle distance as the comparison, three modes behind one parameter. Finds that the detection radius barely matters for coverage while the height threshold decides almost everything — but that read on accuracy the two axes pull the other way, so a coverage-only view overstates the case for a high threshold. The production baseline improves enough under the new candidate stage to overturn version 3's conclusion. Breaks results down by class of aerodrome, which shows the method answering small aerodromes and heliports confidently and wrongly every time. Explains the preprocessing pipeline, each mode, and every metric from first principles.

Detecting departure and destination aerodromes from ADS-B — version 3 · PDF

Finds that six of the seven most frequent departure errors were a scheduled civil airport losing to an adjacent military airfield, and removes that class entirely with a scheduled-service preference. Fixes an out-of-area rule that was pre-empting valid aerodrome matches, reports out-of-area recall and precision separately for the first time, and carries a production patch proposal for `flights.py`.

Detecting departure and destination aerodromes from ADS-B — version 2 · PDF

Corrects two accounting errors in version 1. A large share of flights have an origin or destination outside the ingested area, and scoring those as detection failures understated every method; and the aircraft filter was an allowlist that discarded 47.74% of the fleet to exclude the 8.64% that are rotorcraft. Adds trajectory extrapolation (which fails), the cleaning ablation, the aerodrome-set sweep, an error breakdown, and a measurement of `track_id` quality.

Detecting departure and destination aerodromes from ADS-B — version 1 · PDF

A comparison of seven ADEP/ADES methodologies against APDF ground truth over two three-day samples a year apart. Finds that the production algorithm's altitude-trend logic contributes no accuracy over a nearest-aerodrome rule on the same flights, and that replacing it with explicit endpoint abstention improves departure detection by 8–10 points of coverage — while the same change fails to replicate for arrivals, and is not claimed.